## Lowering of vapour pressure

**1.** (a) $\frac{P^0-Ps}{P^0}=\frac{w\times M}{mW}=143-\frac{0.5\times154}{65\times158}\times143$

$=143-1.03=141.97mm.$

2. **(b) Mole fraction**

**Explanation:**

According to Raoult's law, the partial vapour pressure of a component is directly proportional to its mole fraction in the solution.

$$P_A=X_AP_A^0$$

3. **(b) Raoult's law**

**Explanation:**

Raoult's law states that the relative lowering of vapour pressure is equal to the mole fraction of the solute.

$$\frac{P^0-P}{P^0}=X_B$$

**4.** (d) $\frac{P^0-P_s}{P^0}=\frac{\frac{w}{m}}{\frac{w}{m}+\frac{W}{M}}$ or $0.00713=\frac{71.5/m}{\frac{71.5}{m}+\frac{1000}{18}}$

$m=180$

**5.** (b) $HgI_2$ although insoluble in water but shows complex formation with *KI* and freezing point is decreases.

**6.** (a) For solutions containing non-volatile solutes, the Raoult's law may be stated as at a given temperature, the vapour pressure of a solution containing non-volatile solute is directly proportional to the mole fraction of the solvent.

**7.** (a) Vapour pressure $\propto\frac{1}{\text{Boiling point}}$

**8** **(a)** $P=P^oN_1$

**Explanation:**

According to Raoult's law:

$$P=P^0N_1$$

9 **(c) More than that of water**

**Explanation:**

Methanol is more volatile than water, so adding methanol increases the total vapour pressure.

10 **(d) Saturated vapour pressure**

11 **(a) 1 M $C_{12}H_{22}O_{11}$**

**Explanation:**

Maximum vapour pressure corresponds to minimum number of particles in solution. Sugar does not ionize, while salts produce more particles.

12 **(b) Solute molecules and total molecules in the solution**

**Explanation:**

$$\frac{P^0 - P}{P^0} = X_B = \frac{n_B}{n_A + n_B}$$

13 **(b) Less than vapour pressure of pure water**

**Explanation:**

Adding a solute lowers the vapour pressure of water.

14 **: (c) 0.100**

**Explanation:**

$$X_B = \frac{P^0 - P}{P^0} = \frac{50 - 45}{50} = 0.1$$

15 **(a) 0.549**

**Explanation:**

$$X_{\text{pentane}} = \frac{1}{1+4} = 0.2$$

Partial pressures:

$$P_p = 0.2 \times 440 = 88$$
$$P_h = 0.8 \times 120 = 96$$

Total pressure:

$$P = 88 + 96 = 184$$

Mole fraction in vapour phase:

$$Y_p = \frac{88}{184} = 0.478$$

✅ Correct answer: **(d) 0.478**

16 **(d) 53.5**

**Explanation:**

Moles of benzene:

$$\frac{78}{78} = 1$$

Moles of toluene:

$$\frac{46}{92} = 0.5$$

Mole fraction of benzene:

$$X_B = \frac{1}{1 + 0.5} = \frac{2}{3}$$

Partial pressure:

$$P = 75 \times \frac{2}{3} = 50$$

Correct answer: **(a) 50**

**17** **(b) 0.125 mm Hg**

**Explanation:**

Moles of sucrose:

$$\frac{100}{342} = 0.292$$

Moles of water:

$$\frac{1000}{18} = 55.56$$

Relative lowering:

$$\frac{\Delta P}{P^0} = \frac{0.292}{55.56} = 0.00526$$

$$\Delta P = 23.8 \times 0.00526 \approx 0.125$$

18 **(d) Relative lowering of vapour pressure is directly proportional to original pressure**

19 **(b) Mercury**

**Explanation:**

Mercury is least volatile, hence has the lowest vapour pressure.

20 **(b) Mole fraction of solute**